
Four-bar Linkage Motion

Xie Xuan Zheng

2022/11/30

Contents

1	Formulation of problems	3
1.1	parameters identification	3
1.2	loop equations	3
1.3	Position	4
1.4	velocity	4
1.5	acceleration	6
1.6	geometric constraint	7
2	Procedure of numerical method	8
2.1	Newton-Raphson method	8
2.2	Position analysis using numerical method	9
3	Program	11
3.1	Enviroment	11
3.2	Interface Description	11
3.3	Program structure	12
3.4	Function list	12
4	Discussion	15
4.1	Data analysis	15
4.2	Program improvement	18

Expand the x and y components of loop closure equation of loop2 in equation(1) and substitute the constraints into it.

$$\begin{cases} r_6 \cos(\theta_6) + r_5 \cos(\theta_5) - r_4 \cos(\theta_4) = r_7 \cos(\theta_7) \\ r_6 \sin(\theta_6) + r_5 \sin(\theta_5) - r_4 \sin(\theta_4) = r_7 \sin(\theta_7) \end{cases} \quad (3)$$

Let the input angle is θ_2 . Hence, it is possible to solve $r_3, \theta_3, \theta_5, \theta_6$ via equations (2), (3).

1.3 Position

1.3.1 position of loop1

Rewrite equation (2) into

$$\begin{cases} f_1(r_3, \theta_3) = r_3 \cos(\theta_3) - r_2 \cos(\theta_2) = 0 \\ f_2(r_3, \theta_3) = r_3 \sin(\theta_3) - r_2 \sin(\theta_2) - r_1 = 0 \end{cases} \quad (4)$$

1.3.2 position of loop2

Rewrite equation (3) into

$$\begin{cases} f_3(\theta_5, \theta_6, \theta_3) = r_6 \cos \theta_6 + r_5 \cos \theta_5 - r_4 \cos \theta_3 - r_7 \cos \theta_7 = 0 \\ f_4(\theta_5, \theta_6, \theta_3) = r_6 \sin \theta_6 + r_5 \sin \theta_5 - r_4 \sin \theta_3 - r_7 \sin \theta_7 = 0 \end{cases} \quad (5)$$

However, equations (2), (3) are not linear simultaneous equation. Hence, it must to be solved by numerical method, which will be mentioned in the section2.

1.4 velocity

1.4.1 velocity of loop 1

Take derivative with respect to time once with equation (4)

$$\begin{cases} \dot{r}_3 \cos \theta_3 - r_3 \dot{\theta}_3 \sin \theta_3 = -r_1 \dot{\theta}_2 \sin \theta_2 \\ \dot{r}_3 \sin \theta_3 + r_3 \dot{\theta}_3 \cos \theta_3 = r_2 \dot{\theta}_2 \cos \theta_2 \end{cases}$$

Rewrite into matrix form

$$\begin{aligned}
\begin{pmatrix} \dot{r}_3 \\ \dot{\theta}_3 \end{pmatrix} &= \begin{pmatrix} \cos \theta_3 & -r_3 \sin \theta_3 \\ \sin \theta_3 & r_3 \cos \theta_3 \end{pmatrix}^{-1} \begin{pmatrix} -r_2 & \dot{\theta}_2 & \sin \theta_2 \\ r_1 & \dot{\theta}_2 & \cos \theta_2 \end{pmatrix} \\
&= \frac{1}{r_3} \begin{pmatrix} r_3 \cos \theta_3 & r_3 \sin \theta_3 \\ -\sin \theta_3 & \cos \theta_3 \end{pmatrix} \begin{pmatrix} -r_2 \dot{\theta}_2 & \sin \theta_2 \\ r_1 \dot{\theta}_2 & \cos \theta_2 \end{pmatrix} \\
&= r_1 \dot{\theta}_2 \begin{pmatrix} -\sin \theta_2 \cos \theta_3 + \cos \theta_2 \sin \theta_3 \\ \frac{1}{r_3} \sin \theta_2 \sin \theta_3 + \frac{1}{r_3} \cos \theta_2 \cos \theta_3 \end{pmatrix} \\
&= r_1 \dot{\theta}_2 \begin{pmatrix} \sin(\theta_3 - \theta_2) \\ \frac{1}{r_3} \cos(\theta_3 - \theta_2) \end{pmatrix}
\end{aligned} \tag{6}$$

1.4.2 velocity of loop2

Take derivative with respect to time once with equation (5)

$$\begin{cases} -r_6 \dot{\theta}_6 \sin \theta_6 - r_5 \dot{\theta}_5 \sin \theta_5 + r_4 \dot{\theta}_4 \sin \theta_4 = -r_7 \dot{\theta}_7 \sin \theta_7 \\ r_6 \dot{\theta}_6 \cos \theta_6 + r_5 \dot{\theta}_5 \cos \theta_5 - r_4 \dot{\theta}_4 \cos \theta_4 = r_7 \dot{\theta}_7 \cos \theta_7 \end{cases}$$

Right hand side becomes 0 because θ_7 is constant.

$$\begin{cases} -r_6 \dot{\theta}_6 \sin \theta_6 - r_5 \dot{\theta}_5 \sin \theta_5 = -r_4 \dot{\theta}_4 \sin \theta_4 \\ r_6 \dot{\theta}_6 \cos \theta_6 + r_5 \dot{\theta}_5 \cos \theta_5 = r_4 \dot{\theta}_4 \cos \theta_4 \end{cases}$$

Rewrite into matrix form

$$\begin{aligned}
\begin{pmatrix} \dot{\theta}_5 \\ \dot{\theta}_6 \end{pmatrix} &= \begin{pmatrix} -r_5 \sin \theta_5 & -r_6 \sin \theta_6 \\ r_5 \cos \theta_5 & r_6 \cos \theta_6 \end{pmatrix}^{-1} \begin{pmatrix} -r_4 \dot{\theta}_4 \sin \theta_4 \\ r_4 \dot{\theta}_4 \cos \theta_4 \end{pmatrix} \\
&= \frac{1}{r_5 r_6 (\cos \theta_5 \sin \theta_6 - \sin \theta_5 \cos \theta_6)} \begin{pmatrix} r_6 \cos \theta_6 & r_6 \sin \theta_6 \\ -r_5 \cos \theta_5 & -r_5 \sin \theta_5 \end{pmatrix} \begin{pmatrix} -r_4 \dot{\theta}_4 \sin \theta_4 \\ r_4 \dot{\theta}_4 \cos \theta_4 \end{pmatrix} \\
&= \frac{r_4 \dot{\theta}_4}{r_5 r_6 \sin(\theta_5 - \theta_6)} \begin{pmatrix} r_6 \cos \theta_6 & r_6 \sin \theta_6 \\ -r_5 \cos \theta_5 & -r_5 \sin \theta_5 \end{pmatrix} \begin{pmatrix} -\sin \theta_4 \\ \cos \theta_4 \end{pmatrix} \\
&= \frac{r_4 \dot{\theta}_4}{r_5 r_6 \sin(\theta_5 - \theta_6)} \begin{pmatrix} r_6 (\sin \theta_6 \cos \theta_4 - \cos \theta_6 \sin \theta_4) \\ r_5 (\sin \theta_4 \cos \theta_5 - \sin \theta_5 \cos \theta_4) \end{pmatrix} \\
&= \frac{r_4 \dot{\theta}_4}{r_5 r_6 \sin(\theta_5 - \theta_6)} \begin{pmatrix} r_6 \sin(\theta_6 - \theta_4) \\ r_5 \sin(\theta_4 - \theta_5) \end{pmatrix}
\end{aligned} \tag{7}$$

We can substitute θ_i into equations(6),(7) because they are solved in section 2.2 by numerical method. Finally we can solve $\dot{r}_3$, $\dot{\theta}_3$, $\dot{\theta}_5$, $\dot{\theta}_6$.

1.5 acceleration

1.5.1 acceleration of loop1

Take derivative with respect to time twice with equation (4) $\dot{r}_3, \dot{\theta}_3=\dot{\theta}_4$ is solved in equation(6).Hence, we get

$$\begin{cases} \ddot{r}_3 \cos \theta_3 - \ddot{\theta}_3 r_3 \sin \theta_3 = 2\dot{r}_3 \dot{\theta}_3 \sin \theta_3 + r_3 \left(\dot{\theta}_3 \right)^2 \cos \theta_3 - r_2 \ddot{\theta}_2 \sin \theta_2 - r_2 \left(\dot{\theta}_2 \right)^2 \cos \theta_2 \\ \ddot{r}_3 \sin \theta_3 + \ddot{\theta}_3 r_3 \cos \theta_3 = -2\dot{r}_3 \dot{\theta}_3 \cos \theta_3 + r_3 \left(\dot{\theta}_3 \right)^2 \sin \theta_3 + r_2 \ddot{\theta}_2 \cos \theta_2 - r_2 \left(\dot{\theta}_2 \right)^2 \sin \theta_2 \end{cases}$$

Rewrite into matrix form

$$\begin{aligned} \begin{pmatrix} \ddot{r}_3 \\ \ddot{\theta}_3 \end{pmatrix} &= \begin{pmatrix} \cos \theta_3 & -r_3 \sin \theta_3 \\ \sin \theta_3 & r_3 \cos \theta_3 \end{pmatrix}^{-1} \begin{pmatrix} 2\dot{r}_3 \dot{\theta}_3 \sin \theta_3 + r_3 \left(\dot{\theta}_3 \right)^2 \cos \theta_3 - r_2 \ddot{\theta}_2 \sin \theta_2 - r_2 \left(\dot{\theta}_2 \right)^2 \cos \theta_2 \\ -2\dot{r}_3 \dot{\theta}_3 \cos \theta_3 + r_3 \left(\dot{\theta}_3 \right)^2 \sin \theta_3 + r_2 \ddot{\theta}_2 \cos \theta_2 - r_2 \left(\dot{\theta}_2 \right)^2 \sin \theta_2 \end{pmatrix} \\ &= \frac{1}{r_3} \begin{pmatrix} r_3 \cos \theta_3 & r_3 \sin \theta_3 \\ -\sin \theta_3 & \cos \theta_3 \end{pmatrix} \begin{pmatrix} 2\dot{r}_3 \dot{\theta}_3 \sin \theta_3 + r_3 \left(\dot{\theta}_3 \right)^2 \cos \theta_3 - r_1 \ddot{\theta}_2 \sin \theta_2 - r_2 \left(\dot{\theta}_2 \right)^2 \cos \theta_2 \\ -2\dot{r}_3 \dot{\theta}_3 \cos \theta_3 + r_3 \left(\dot{\theta}_3 \right)^2 \sin \theta_3 + r_2 \ddot{\theta}_2 \cos \theta_2 - r_2 \left(\dot{\theta}_2 \right)^2 \sin \theta_2 \end{pmatrix} \\ &= \begin{pmatrix} r_3 \left(\dot{\theta}_3 \right)^2 + r_2 \ddot{\theta}_2 \sin (\theta_3 - \theta_2) - r_2 \left(\dot{\theta}_2 \right)^2 \cos (\theta_3 - \theta_2) \\ -2\frac{\dot{r}_3}{r_3} \dot{\theta}_3 + \frac{r_1}{r_3} \ddot{\theta}_2 \sin (\theta_2 + \theta_3) + \frac{r_2}{r_3} \left(\dot{\theta}_2 \right)^2 \sin (\theta_3 - \theta_2) \end{pmatrix} \end{aligned} \tag{8}$$

1.5.2 acceleration of loop2

Take derivative with respect to time twice with equation (5). $\dot{\theta}_3=\dot{\theta}_4$ is solved in equation(6).Hence we get

$$\begin{cases} \ddot{\theta}_6 (-r_6 \sin \theta_6) + \ddot{\theta}_5 (-r_5 \sin \theta_5) = r_6 \left(\dot{\theta}_6 \right)^2 \cos \theta_6 + r_5 \left(\dot{\theta}_5 \right)^2 \cos \theta_5 - r_4 \ddot{\theta}_4 \cos \theta_4 + r_4 \left(\dot{\theta}_4 \right)^2 \sin \theta_4 \\ \ddot{\theta}_6 (r_6 \cos \theta_6) + \ddot{\theta}_5 (r_5 \cos \theta_5) = r_6 \left(\dot{\theta}_6 \right)^2 \sin \theta_6 + r_5 \left(\dot{\theta}_5 \right)^2 \sin \theta_5 + r_4 \ddot{\theta}_4 \sin \theta_4 + r_4 \left(\dot{\theta}_4 \right)^2 \cos \theta_4 \end{cases}$$

Rewrite into matrix form

$$\begin{aligned}
\begin{pmatrix} \ddot{\theta}_5 \\ \ddot{\theta}_6 \end{pmatrix} &= \begin{pmatrix} -r_5 \sin \theta_5 & -r_6 \sin \theta_6 \\ r_5 \cos \theta_5 & r_6 \cos \theta_6 \end{pmatrix}^{-1} \begin{pmatrix} r_6 \left(\dot{\theta}_6 \right)^2 \cos \theta_6 + r_5 \left(\dot{\theta}_5 \right)^2 \cos \theta_5 - r_4 \ddot{\theta}_4 \cos \theta_4 + r_4 \left(\dot{\theta}_4 \right)^2 \sin \theta_4 \\ r_6 \left(\dot{\theta}_6 \right)^2 \sin \theta_6 + r_5 \left(\dot{\theta}_5 \right)^2 \sin \theta_5 + r_4 \ddot{\theta}_4 \sin \theta_4 + r_4 \left(\dot{\theta}_4 \right)^2 \cos \theta_4 \end{pmatrix} \\
&= \frac{1}{r_5 r_6 \sin(\theta_5 - \theta_6)} \begin{pmatrix} r_6 \cos \theta_6 & r_6 \sin \theta_6 \\ -r_5 \cos \theta_5 & -r_5 \sin \theta_5 \end{pmatrix} \begin{pmatrix} r_6 \left(\dot{\theta}_6 \right)^2 \cos \theta_6 + r_5 \left(\dot{\theta}_5 \right)^2 \cos \theta_5 - r_4 \ddot{\theta}_4 \cos \theta_4 + r_4 \left(\dot{\theta}_4 \right)^2 \sin \theta_4 \\ r_6 \left(\dot{\theta}_6 \right)^2 \sin \theta_6 + r_5 \left(\dot{\theta}_5 \right)^2 \sin \theta_5 + r_4 \ddot{\theta}_4 \sin \theta_4 + r_4 \left(\dot{\theta}_4 \right)^2 \cos \theta_4 \end{pmatrix} \\
&= \frac{1}{\sin(\theta_6 - \theta_5)} \begin{pmatrix} \frac{r_6}{r_5} \left(\dot{\theta}_6 \right)^2 + \left(\dot{\theta}_5 \right)^2 \sin(\theta_6 + \theta_5) + \frac{r_4}{r_5} \ddot{\theta}_4 \sin(\theta_6 - \theta_4) + \frac{r_4}{r_5} \left(\dot{\theta}_4 \right)^2 \sin(\theta_6 + \theta_4) \\ - \left(\dot{\theta}_6 \right)^2 \cos(\theta_6 - \theta_5) - \frac{r_5}{r_6} \left(\dot{\theta}_5 \right)^2 + \frac{r_4}{r_6} \ddot{\theta}_4 \cos(\theta_5 + \theta_4) - \frac{r_4}{r_6} \left(\dot{\theta}_4 \right)^2 \sin(\theta_5 + \theta_4) \end{pmatrix} \tag{9}
\end{aligned}$$

θ_i are solved in section 2.2 using numerical method. $\dot{\theta}_i$ are solved in section 1.4. Hence, we can substitute the solution of θ_i , $\dot{\theta}_i$ into equations(8), (9). Finally we obtain $\ddot{r}_3$, $\ddot{\theta}_3$, $\ddot{\theta}_5$, $\ddot{\theta}_6$.

1.6 geometric constraint

1.6.1 geometric constraint of loop1

θ_2 is input angle with respect to loop1. Loop1 should follow triangle inequality and the slider3 should not exceed link4. Hence below is the constraint of loop1.

$$\left\{ \begin{array}{l} |\vec{r}_3| \leq |\vec{r}_4| \\ |\vec{r}_1| + |\vec{r}_3| \leq |\vec{r}_2| \text{ and } |\vec{r}_1| + |\vec{r}_2| \leq |\vec{r}_3| \text{ and } |\vec{r}_2| + |\vec{r}_3| \leq |\vec{r}_1| \end{array} \right.$$

1.6.2 geometric constraint of loop2

$\theta_3 = \theta_4$ and they are input with respect to loop2. We can obtain the following equation by equation(1)

$$\begin{aligned}
r_5^2 &= (r_4 \cos \theta_4 + r_7 \cos \theta_7 - r_6 \cos \theta_6)^2 + (r_4 \sin \theta_4 + r_7 \sin \theta_7 - r_6 \sin \theta_6)^2 \\
&= r_4^2 \cos^2 \theta_4 + r_7^2 \cos^2 \theta_7 + r_6^2 \cos^2 \theta_6 + 2r_4 r_7 \cos \theta_4 \cos \theta_7 - 2r_4 r_6 \cos \theta_4 \cos \theta_6 - 2r_7 r_6 \cos \theta_7 \cos \theta_6 \\
&\quad + r_4^2 \sin^2 \theta_4 + r_7^2 \sin^2 \theta_7 + r_6^2 \sin^2 \theta_6 + 2r_4 r_7 \sin \theta_4 \sin \theta_7 - 2r_4 r_6 \sin \theta_4 \sin \theta_6 - 2r_7 r_6 \sin \theta_7 \sin \theta_6 \\
&= r_4^2 + r_6^2 + r_7^2 + 2r_4 r_7 (\cos \theta_4 \cos \theta_7 + \sin \theta_4 \sin \theta_7) - 2r_4 r_6 (\cos \theta_4 \cos \theta_6 + \sin \theta_4 \sin \theta_6) \\
&\quad - 2r_7 r_6 (\cos \theta_7 \cos \theta_6 + \sin \theta_7 \sin \theta_6)
\end{aligned}$$

Rewrite the above equation into $A \cos(\theta_6) + B \sin(\theta_6) + C = 0$, $A^2 + B^2 \geq C^2$

$$A = -2r_4 r_6 \cos \theta_4 - 2r_7 r_6 \cos \theta_7$$

$$B = -2r_4 r_6 \sin \theta_4 - 2r_7 r_6 \sin \theta_7$$

$$C = r_4^2 + r_6^2 + r_7^2 - r_5^2 + 2r_4 r_7 (\cos \theta_4 \cos \theta_7 + \sin \theta_4 \sin \theta_7)$$

Finally, we can justify if a geometry of loop2 is valid with $A^2 + B^2 \geq C^2$.

2 Procedure of numerical method

2.1 Newton-Raphson method

The following equation is taking linear approximation using Taylor expansion.

$$f_i(\hat{\theta} + \Delta\theta) = f_i(\hat{\theta}) + \sum_{k=1}^n \left(\frac{\partial f_i(\theta)}{\partial \theta_k} \right)_{\hat{\theta}} \Delta\theta_k$$

If we need to find the roots of a function, then the right hand side of the equation is 0.

$$f_i(\hat{\theta} + \Delta\theta) = f_i(\hat{\theta}) + \sum_{k=1}^n \left(\frac{\partial f_i(\theta)}{\partial \theta_k} \right)_{\hat{\theta}} \Delta\theta_k = 0$$

$$f_i(\hat{\theta}) = - \sum_{k=1}^n \left(\frac{\partial f_i(\theta)}{\partial \theta_k} \right)_{\hat{\theta}} \Delta\theta_k$$

Finally, we use Newton-Raphson method to find the solutions iteratively after defining a proper error ϵ .

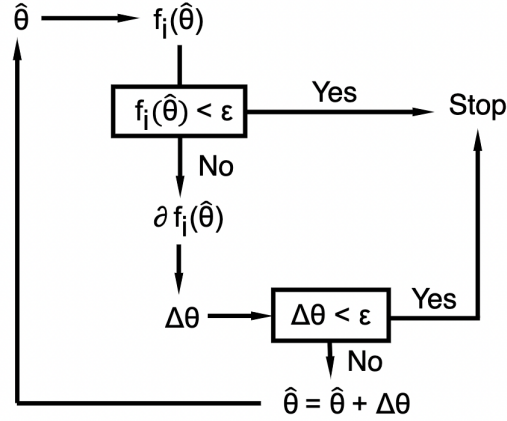


Figure 2: Newton-Raphson method

2.2 Position analysis using numerical method

2.2.1 Position analysis using numerical method for loop1

Taking Taylor expansion to the first function of equation (4)

$$\begin{aligned} f_1(\hat{r}_3, \hat{\theta}_3) + \left(\frac{\partial}{\partial r_3} f_1(\hat{\theta}) \right)_{\hat{r}_3} \Delta r_3 + \left. \frac{\partial}{\partial \theta_3} f_1(\hat{\theta}) \right|_{\theta_3} \Delta \theta_3 &= 0 \\ \Rightarrow \cos \hat{\theta}_3 \Delta r_3 - \hat{r}_3 \sin \hat{\theta}_3 \Delta \theta_3 &= -f_1(\hat{r}_3, \hat{\theta}_3) \end{aligned}$$

Taking Taylor expansion to the second function of equation (4)

$$\begin{aligned} f_2(\hat{r}_3, \hat{\theta}_3) + \left(\frac{\partial}{\partial r_3} f_2(\hat{\theta}) \right)_{\hat{r}_3} \Delta r_3 + \left. \frac{\partial}{\partial \theta_3} f_2(\hat{\theta}) \right|_{\theta_3} \Delta \theta_3 &= 0 \\ \Rightarrow \sin \hat{\theta}_3 \Delta r_3 - \hat{r}_3 \cos \hat{\theta}_3 \Delta \theta_3 &= -f_2(\hat{r}_3, \hat{\theta}_3) \end{aligned}$$

Rewrite into matrix form

$$\begin{aligned} \begin{pmatrix} \Delta r_3 \\ \Delta \theta_3 \end{pmatrix} &= \begin{pmatrix} -\cos \hat{\theta}_3 & \hat{r}_3 \sin \hat{\theta}_3 \\ -\sin \hat{\theta}_3 & -\hat{r}_3 \cos \hat{\theta}_3 \end{pmatrix}^{-1} \begin{pmatrix} f_1(\hat{r}_3, \hat{\theta}_3) \\ f_2(\hat{r}_3, \hat{\theta}_3) \end{pmatrix} \\ &= \frac{1}{\hat{r}_3} \begin{pmatrix} -\hat{r}_3 \cos \hat{\theta}_3 & -\hat{r}_3 \sin \hat{\theta}_3 \\ \sin \hat{\theta}_3 & -\cos \hat{\theta}_3 \end{pmatrix} \begin{pmatrix} f_1(\hat{r}_3, \hat{\theta}_3) \\ f_2(\hat{r}_3, \hat{\theta}_3) \end{pmatrix} \\ &= \begin{pmatrix} -\cos \hat{\theta}_3 f_1(\hat{r}_3, \hat{\theta}_3) - \sin \hat{\theta}_3 f_2(\hat{r}_3, \hat{\theta}_3) \\ \frac{1}{\hat{r}_3} \sin \hat{\theta}_3 f_1(\hat{r}_3, \hat{\theta}_3) - \frac{1}{\hat{r}_3} \cos \hat{\theta}_3 f_2(\hat{r}_3, \hat{\theta}_3) \end{pmatrix} \end{aligned} \quad (10)$$

Substitute $\theta_3 = \theta_4$ which is solved in equation (2) into above equation.

2.2.2 Position analysis using numerical method for loop2

Take Taylor expansion of the first function of equation(5)

$$\begin{aligned} f_3(\hat{r}_5, \hat{\theta}_6) + \left. \frac{\partial}{\partial \theta_5} f_3(\hat{\theta}) \right|_{\hat{\theta}_5} \Delta \theta_5 + \left. \frac{\partial}{\partial \theta_6} f_3(\hat{\theta}) \right|_{\theta_6} \Delta \theta_6 &= 0 \\ \Rightarrow -r_5 \sin \hat{\theta}_5 \Delta \theta_5 - r_6 \sin \hat{\theta}_6 \Delta \theta_6 &= -f_3(\hat{\theta}_5, \hat{\theta}_6) \end{aligned}$$

Take Taylor expansion of the second function of equation(5)

$$\begin{aligned} f_4(\hat{r}_5, \hat{\theta}_6) + \left. \frac{\partial}{\partial \theta_5} f_4(\hat{\theta}) \right|_{\hat{\theta}_5} \Delta \theta_5 + \left. \frac{\partial}{\partial \theta_6} f_4(\hat{\theta}) \right|_{\theta_6} \Delta \theta_6 &= 0 \\ \Rightarrow r_5 \cos \hat{\theta}_5 \Delta \theta_5 + r_6 \cos \hat{\theta}_6 \Delta \theta_6 &= -f_4(\hat{\theta}_5, \hat{\theta}_6) \end{aligned}$$

Rewrite into matrix form

$$\begin{aligned}
\begin{pmatrix} \Delta\theta_5 \\ \Delta\theta_6 \end{pmatrix} &= \begin{pmatrix} r_5 \sin \hat{\theta}_5 & r_6 \sin \hat{\theta}_6 \\ -r_5 \cos \hat{\theta}_5 & -r_6 \cos \hat{\theta}_6 \end{pmatrix}^{-1} \begin{pmatrix} f_3(\hat{\theta}_5, \hat{\theta}_6) \\ f_4(\hat{\theta}_5, \hat{\theta}_6) \end{pmatrix} \\
&= \frac{1}{r_5 r_6 (\sin \hat{\theta}_6 \cos \hat{\theta}_5 - \cos \hat{\theta}_6 \sin \hat{\theta}_5)} \begin{pmatrix} -r_6 \cos \hat{\theta}_6 & -r_6 \sin \hat{\theta}_6 \\ r_5 \cos \hat{\theta}_5 & r_5 \sin \hat{\theta}_5 \end{pmatrix} \begin{pmatrix} f_3(\hat{\theta}_5, \hat{\theta}_6) \\ f_4(\hat{\theta}_5, \hat{\theta}_6) \end{pmatrix} \\
&= \frac{1}{\sin(\hat{\theta}_6 - \hat{\theta}_5)} \begin{pmatrix} -\frac{\cos \hat{\theta}_6}{r_5} & -\frac{\sin \hat{\theta}_6}{r_5} \\ \frac{\cos \hat{\theta}_5}{r_6} & \frac{\sin \hat{\theta}_5}{r_6} \end{pmatrix} \begin{pmatrix} f_3(\hat{\theta}_5, \hat{\theta}_6) \\ f_4(\hat{\theta}_5, \hat{\theta}_6) \end{pmatrix} \\
&= \frac{1}{\sin(\hat{\theta}_6 - \hat{\theta}_5)} \begin{pmatrix} -\frac{\cos \hat{\theta}_6}{r_5} f_3(\hat{\theta}_5, \hat{\theta}_6) - \frac{\sin \hat{\theta}_6}{r_5} f_4(\hat{\theta}_5, \hat{\theta}_6) \\ \frac{\cos \hat{\theta}_5}{r_6} f_3(\hat{\theta}_5, \hat{\theta}_6) + \frac{\sin \hat{\theta}_5}{r_6} f_4(\hat{\theta}_5, \hat{\theta}_6) \end{pmatrix}
\end{aligned} \tag{11}$$

By Newton-Raphson method, we can solve $r_3, \theta_3, \theta_5, \theta_6$ after obtaining equation(10), (11).

3 Program

3.1 Enviroment

In order to achieve the goal of UI and numerical computing, Python is chosen to implement because of its complete module; Besides memory and algorithm is neglectable due to no size changing with input data. The program is runned on the enviroment of Pthon 3.10.8 and it is require for modules of customtkinter and matplotlib. If the program hasn't been compiled succesfully, then it should use pip to install in terminal.

cammand: `pip install tkinter pip install matplotlib`

3.2 Interface Description

The user interface is shown in figure3. It is divided into three parts, which is left top, right top and bottom. Left top has a canvas that display the animation. Right top has eight entries and two button. Entries can type into number of custom parameter. Display button can display the animation after being clicked. GraphIt button can plotted the charts in another frame. If the number is invalid, then UI will show error message on canvas. Bottom is a scrollable table of the data.

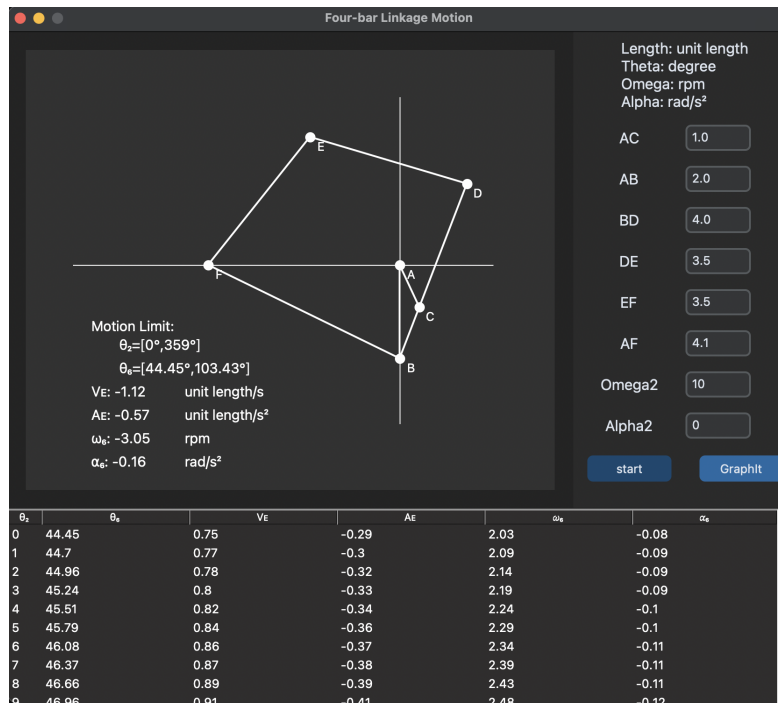


Figure 3: Interface

3.3 Program structure

As shown in figure4, the program is divided into three parts. main.py is the main function, which instantiate the App object. Kernel.py is in charge of calculating the animation data with respect to input data in entries. Interface.py is in charge of UI layout , animation displaying and plotting data.

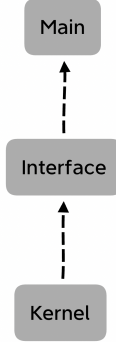


Figure 4: program structure

3.4 Function list

3.4.1 Kernel

1. `__init__()`
 The function is the constructor of `animation_kernel`. Dictionary is used due to the multi variable.
 - (a) The dictionary data is mapped by the key θ_2 and is mapped to all of the solution. The key "length" is mapped to the value r_i . The key "theta" is mapped to the value θ_i . The key "omega" is mapped to the value $\dot{\theta}_i$. The key "alpha" is mapped to $\ddot{\theta}_i$. The key "r3_vel" is mapped to $\dot{r}_3$. The key "r3_acc" is mapped to the value $\ddot{r}_3$. The key "motion_limit_loop1" is mapped to the motion limit of loop1.
 - (b) Dictionary animation is mapped by the key θ_2 . It use the data structure as $\{\theta_2: [\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6, \theta_7], \omega_6, v_6, \alpha_6, a_6\}$
 - (c) Initial guess is also executed in the constructor. The initial guess is $r_3 = \overline{AB} - \overline{AC}$, $\theta_3 = 190^\circ$, $\theta_4 = 190^\circ$, $\theta_5 = 120^\circ$, $\theta_6 = 100^\circ$
 - (d) It will call `find_motion_limit_loop2()` to find the motion limit of loop2 after create an `animation_kernel`.
2. `find_motion_limit_loop2()`
 The function is aim to calculate the motion limit of loop2 by the result of section 1.6.2. The method is let θ_3 be the input angle of loop2. If there are two root to $A^2 + B^2 - C^2 = 0$, then return True, which means that loop2 has motion and return False otherwise.

3. `find_theta()`
 The function id aim to find $r_3, \theta_3, \theta_5, \theta_6$ with respect to an input θ_2 . Caller will modified θ_2 first and then call this function to implement calculation. There are two possible error of this function
 - (a) zero division error
 - (b) time limit exceed

Solution Let max iterative times being 100 and let the solution with respect to θ_2 be the solution with respect to $\theta_2 - 1$.
4. `find_omega()`
 The function is aim to calculate omega with respect to an input θ_2 using the equations obtained in section 1.4. It is called after `find_theta()` and substitute θ_i into itself to find θ_i . The solution with respect to θ_2 will be the solution respect to $\theta_2 - 1$ in order to prevent zero division error.
5. `find_alpha()`
 The function is aimed to calculate $\ddot{\theta}_i$ by the result in section 1.5. It should be called after `find_theta()` and `find_omega()` and substitute $\theta_i, \dot{\theta}_i$ into itself. The solution with respect to θ_2 will be the solution respect to $\theta_2 - 1$ in order to prevent zero division error.
6. `find_motion_limit_loop1()`
 The function is aim to calculate the motion limit of loop1. Due to the motion limit of loop2 is obtained while the object is created, let the θ_2 iterate from 0° to 360° . If there is a solution set, then append the set θ_i into animation in list form first.
7. `get_animation()`
 The function is aim to find the animation with respect to the input data. If the input data allows r_3 to rotate for one circle, then the animation is calculate from 230° and calculate from the start of motion limit otherwise. Consider the case of angular acceleration, it should update θ_2 via $\omega = \sqrt{\omega_0^2 + 2\alpha(\theta - \theta_{start})}$ in every iteration. In the same time, calling `find_omega()` and `find_alpha()` to obtain ω_i and α_i . Finally, append $\omega_6, v_6, \alpha_6, a_6$ into animation.

3.4.2 Interface

1. `__init__()`
 The function is the consteuctor of App object. This object inherit from customer.CTK object, hence the UI is shown after creating an instance. Also, the function will create widgets of UI.
2. `UIlayout()`
 The function is aim to put the widgets on the main frame and bind input events to entries at the same time, such that the obtaining a new animation kernel while typing values.
3. `create_animation_kernel()`
 The function is aim to get the value in entries and return an instance of `animation_kernel`.

4. `change_btn_state()`
The function is aim to change the state of animation button. The state \normal” means that the animation is ready to being display. The state \active” means that the animation is ready to being paused.
5. `draw_animation()`
The function is aim to draw the animation with respect to an input value as key. First of all, the function will obtain the coordinate of point A,B,C,D,E,F. $A=(0,0)$, $B=(0,-\overline{AB})$, $F=(-\overline{AF},0)$, $C=(\overline{AC}\cos(\theta_2), \overline{AC}\sin(\theta_2))$, $D=(B_x+\overline{BD}\cos(\theta_4), B_y+\overline{BD}\sin(\theta_4))$, $E=(F_x+\overline{EF}\cos(\theta_6), F_y+\overline{EF}\sin(\theta_6))$. After that, adding offset of x-axis and y-axis of origin in canvas and add the line between each point.
6. `init_animation()`
The function is aim to init the animation after the new data is typed into entries. By the binding events, it will be the call back function.
 - (a) If the animation is playing, then call `change_btn_state()` to stop the animation.
 - (b) Calling `create_animation_kernel()` to instantiate `animation_kernel`.
 - (c) Using `get_animation()` method of `animation_kernel` to get data of animation.
 - (d) Using `get_graph()` method of `animation_kernel` to get data of graph.
 - (e) Using `draw_animation()` method to show the preview of animation.

If it cannot obtain animation, then ”invalid input” will be shown on the canvas.
7. `start_animation()`
The function is aim to display the animation with respect to input data. After calling `init_animation()`, the dictionary animation is obtained.
 - (a) Pushing every key in animation to a double ended queue to map the animation by θ_2 . Also, the rotate method of double ended queue can achieve the goal of replaying the animation.
 - (b) Setting the animation speed as $\frac{1}{\omega_2}$ second to update the animation of next θ_2 .

In order to prevent stack overflow from excessively updating, set the animation speed limit in 1 ms and show ”too fast” error on canvas.
8. `on_animation_btn_click()`
The function is the callback function of entries.
 - (a) Changing the state of display button
 - (b) Displaying the animation

4 Discussion

4.1 Data analysis

4.1.1 input data1

The first data set is $\omega_2 = 10 \text{ rpm}$, $\alpha_2 = 0 \text{ rad/s}^2$, $\overline{AB} = 2$, $\overline{AC} = 1$, $\overline{BD} = 4$, $\overline{DE} = 3.5$, $\overline{EF} = 3.5$, $\overline{AF} = 4.1$. The data is plotted in figure5. As r_3 can rotate for one circle, the animation is calculated from 230° .

1. discussion of motion limit of link 6
 $\theta_2 \in [0^\circ, 359^\circ]$, $\theta_6 \in [44.45^\circ, 103.43^\circ]$
2. discussion of ω_6
 ω_6 is negative and link6 rotate in CW while $\theta_2 \in [210^\circ, 330^\circ]$. ω_6 is positive and link6 rotate in CCW while $\theta_2 \in [331^\circ, 360^\circ]$, $\theta_2 \in [0^\circ, 209^\circ]$.
 $|\omega_6|_{max} = 4.97 \text{ rpm}$ while $\theta_2 \in [101^\circ, 133^\circ]$, $\theta_2 \in [255^\circ, 266^\circ]$.
3. discussion of α_6
 α_6 is negative and link 6 accelerate in CW while $\theta_2 \in [0^\circ, 164^\circ]$. α_6 is positive and link 6 accelerate in CCW while $\theta_2 \in [164^\circ, 242^\circ]$. $|\alpha_6|_{max} = 3.07 \text{ rad/s}^2$ while $\theta_2 \in [81^\circ, 85^\circ]$.

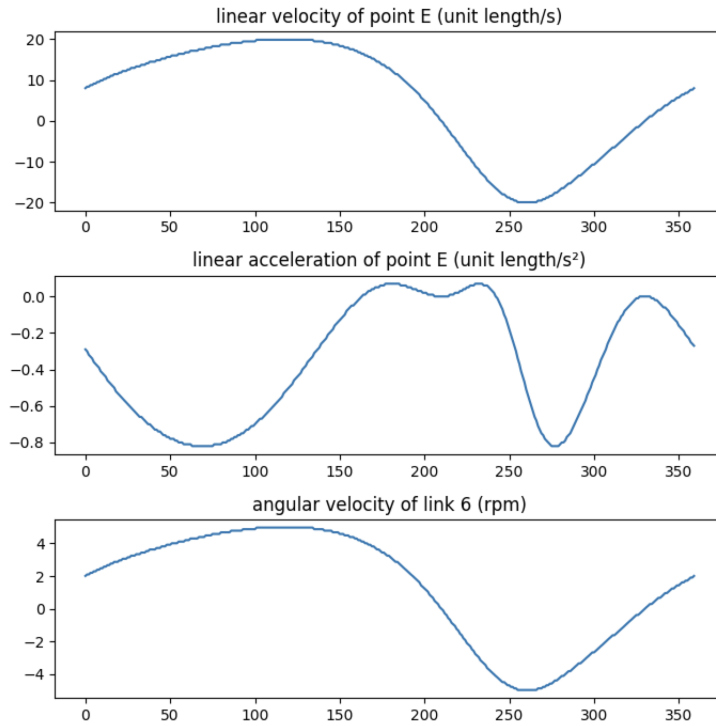


Figure 5: data1

4.1.2 input data 2

The data set2 is $\omega_2 = 0 \text{ rpm}$, $\alpha_2 = 2 \text{ rad/s}^2$, $\overline{AB} = 2$, $\overline{AC} = 1$, $\overline{BD} = 4$, $\overline{DE} = 3.5$, $\overline{EF} = 3.5$, $\overline{AF} = 4.1$ As r_3 can rotate for one circle, the animation is calculated from 230° . The break point in figure6 is result from the angular acceleration is not equal to zero.

1. discussion of motion limit of link 6
 $\theta_2 \in [0^\circ, 359^\circ]$, $\theta_6 \in [44.45^\circ, 103.43^\circ]$
2. discussion of ω_6
 ω_6 is negative and link6 rotate in CW while $\theta_2 \in [211^\circ, 329^\circ]$. ω_6 is positive and link6 rotate in CCW while $\theta_2 \in [330^\circ, 360^\circ]$, $\theta_2 \in [0^\circ, 210^\circ]$.
 $|\omega_6|_{max} = 20.06 \text{ rpm}$ while $\theta_2 \in [135^\circ, 139^\circ]$.
3. discussion of α_6
 α_6 is negative and link 6 accelerate in CW while $\theta_2 \in [0^\circ, 164^\circ]$, $\theta_2 \in [243^\circ, 360^\circ]$. α_6 is positive and link 6 accelerate in CCW while $\theta_2 \in [164^\circ, 242^\circ]$. $|\alpha_6|_{max} = 3.07 \text{ rad/s}^2$ while $\theta_2 \in [81^\circ, 85^\circ]$.

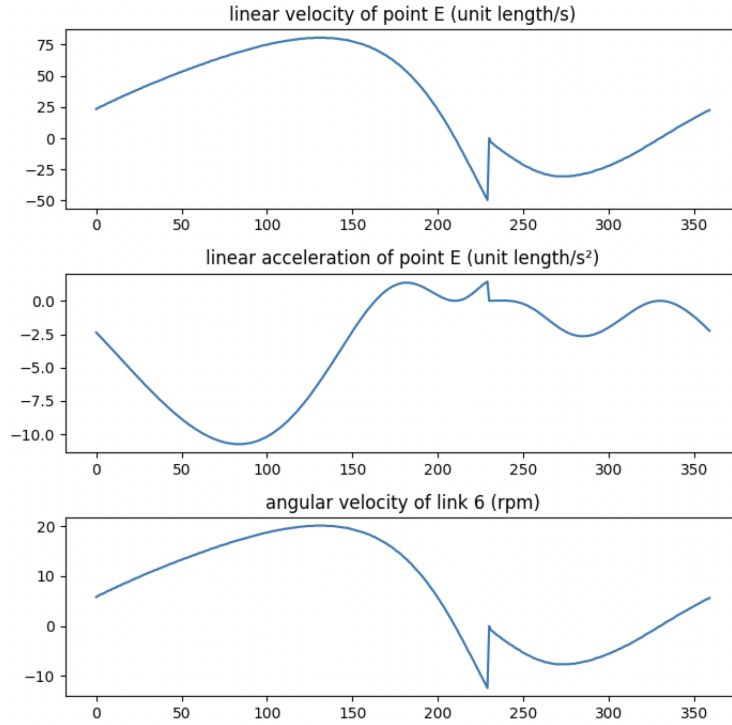


Figure 6: data2

4.1.3 input data 3

The input data3 is $\omega_2 = 10 \text{ rpm}$, $\alpha_2 = 2 \text{ rad/s}^2$, $\overline{AB} = 2$, $\overline{AC} = 1$, $\overline{BD} = 4$, $\overline{DE} = 1.5$, $\overline{EF} = 1.5$, $\overline{AF} = 4.1$ As r_3 cannot rotate for one circle, animation is calculated from the start of motion limit.

1. discussion of motion limit of link 6
 $\theta_2 \in [168^\circ, 240^\circ]$, $\theta_6 \in [39.97^\circ, 66.31^\circ]$
2. discussion of ω_6
 ω_6 is negative and link6 rotate in CW while $\theta_2 \in [210^\circ, 240^\circ]$. ω_6 is positive and link6 rotate in CCW while $\theta_2 \in [168^\circ, 209^\circ]$, $\theta_2 \in [330^\circ, 360^\circ]$.
 $|\omega_6|_{max} = 105 \text{ rpm}$ while $\theta_2 = 240^\circ$.
3. discussion of α_6
 α_6 is negative and link 6 accelerate in CW while $\theta_2 \in [168^\circ, 240^\circ]$. $|\alpha_6|_{max} = 2360 \text{ rad/s}^2$ while $\theta_2 = 240^\circ$.

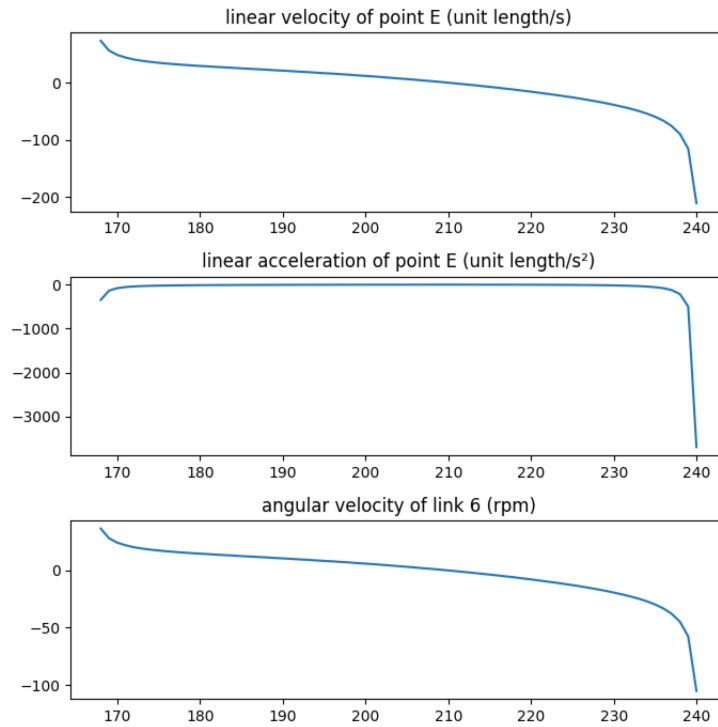


Figure 7: data3

4.2 Program improvement

4.2.1 interface

1. The entries will focus out while plotting, making the entries cannot type successively. Using GrapGIt button to avoid this problem.
2. If the input data does not allow r_3 to rotate for one circle, motion limit may get a little bit error because the motion limit is calculated by the angle between x-axis and r_3 in CCW.
3. In order to replay animation, using double ended queue to implement. However, there might be a proper way to design index to decrease CPU time.

4.2.2 Kernel

1. We use the result by the method in section 1.6.2 and iterate per 1° . If we want to find motion limit more precisely, then we should use some root finding numerical method like bisection method.
2. Without constructing a function call flow chart, there might be a more efficient step to calculate animation.
3. find_theta find_omega find_alpha make the solution with respect to θ_2 as the solution respect to $\theta_2 - 1$ in order to prevent zero division error. It may cause deviation.